

Deutsche Akkreditierungsstelle

Annex to the Accreditation Certificate D-PL-20469-01-02 according to DIN EN ISO/IEC 17025:2018

Valid from: 25.08.2025

Date of issue: 25.08.2025

This annex is part of the Accreditation Certificate D-PL-20469-01-00.

Holder of the Accreditation Certificate:

muva kempten GmbH
Ignaz-Kiechle-Straße 20-22, 87437 Kempten

with the location

muva kempten GmbH
Ignaz-Kiechle-Straße 20-22, 87437 Kempten

The testing laboratory meets the requirements of DIN EN ISO/IEC 17025:2018 to carry out the conformity assessment activities listed in this annex. The testing laboratory meets additional legal and normative requirements, if applicable, including those in relevant sectoral schemes, provided that these are explicitly confirmed below.

The management system requirements of DIN EN ISO/IEC 17025 are written in the language relevant to the operations of testing laboratories and they conform to the principles of DIN EN ISO 9001.

*This annex to the certificate was issued by the Deutsche Akkreditierungsstelle GmbH (DAkkS) and is digitally sealed.
This annex to the certificate is only valid together with the written accreditation certificate and reflects the status as indicated by the date of issue. The current status of any valid and surveyed accreditation can be found in the directory of accredited bodies maintained by Deutsche Akkreditierungsstelle GmbH (www.dakks.de).*

Abbreviations used: see last page

page 1 of 17

This document is a translation. The definitive version is the original German annex to the accreditation certificate.

Annex to the Accreditation Certificate D-PL-20469-01-02

Tests in the fields:

physical, physico-chemical, sensory and microbiological analysis of water (drinking water, industrial water, surface water, process water, raw water and groundwater, swimming pool and bathing pool water and water from small bathing ponds);

Sampling of swimming pool and bathing pool water,

microbiological and selected chemical analysis in accordance with the German Drinking Water Ordinance, sampling of raw and drinking water;

sampling and microbiological analysis of industrial water in accordance with Section 3 (8) 42nd BImSchV (Federal Immission Control Ordinance);

Specialist module for water

Flexible Scope of Accreditation:

The testing laboratory is permitted to use standardised or equivalent test methods listed here with different issue dates without being required to prior inform and obtain approval from DAkkS (flexibilization according to category A).

This does not apply to the specialist module for water.

The testing laboratory has an up-to-date list of all test methods within the flexible scope of accreditation. The list is publicly available on the website of the testing laboratory.

1 Analysis of water (drinking water, industrial water, surface water, process water, raw water and groundwater, swimming pool and bathing pool water and water from small bathing ponds)

1.1 Sampling and sample pretreatment

DIN EN ISO 19458 (K 19) 2006-12	Water quality – Sampling for microbiological analysis
DIN 19643-1 2023-06	Treatment of swimming pool and bathing pool water – Part 1: General requirements (Restriction: <i>here only 14.2, Sampling points and sampling</i>)
UBA Recommendation 2018-12	Systemic analysis of drinking water installations for legionella in accordance with the German Drinking Water Ordinance – Sampling, examination and indication of the result
UBA Recommendation 2018-12	Assessment of the quality of drinking water with respect to the parameters lead, copper and nickel (“sampling recommendation”)

Valid from: 25.08.2025

Date of issue: 25.08.2025

page 2 of 17

This document is a translation. The definitive version is the original German annex to the accreditation certificate.

Annex to the Accreditation Certificate D-PL-20469-01-02

1.2 Sensory and visual examination

DIN EN 1622 (B 3) 2006-10	Water quality – Determination of the threshold odour number (TON) and threshold flavour number (TFN)
------------------------------	--

1.3 Physical and physico-chemical parameters

DIN EN ISO 7887 (C 1) 2012-04	Water quality – Examination and determination of colour
DIN 38404-C 3 2005-07	Determination of absorption in the range of UV radiation, spectral absorption coefficient
DIN 38404-C 4 1976-12	Determination of temperature
DIN EN ISO 10523 (C 5) 2012-04	Water quality – Determination of pH
DIN EN 27888 (C 8) 1993-11	Water quality – Determination of electrical conductivity
DIN 38404-C 10 2012-12	Calculation of the calcit saturation of water
DIN EN ISO 7027-1 (C 21) 2016-11	Water quality – Determination of turbidity – Part 1: Quantitative method
DIN EN ISO 7027-2 (C 22) 2019-06	Water quality – Determination of turbidity – Part 2: Semi-quantitative methods for the assessment of transparency of waters (Restriction: <i>here only method 1 a) measurement of visual range using the transparency testing tube and 1 b) measurement of visual range in the upper water layers using the transparency testing disc</i>)

1.4 Anions

DEV D 8 1971	Determination of the hydrogen carbonate ion (hydrogen carbonate hardness)
DIN 38405-D 9 2011-09	Photometric determination of nitrate
DIN EN 26777 (D 10) 1993-04	Water quality; determination of nitrite; molecular absorption spectrometric method

Valid from: 25.08.2025

Date of issue: 25.08.2025

page 3 of 17

This document is a translation. The definitive version is the original German annex to the accreditation certificate.

Annex to the Accreditation Certificate D-PL-20469-01-02

DIN EN ISO 6878 (D 11) 2004-09	Water quality – Determination of phosphorus – Ammonium molybdate photometric method
DIN 38405-D 13 2011-04	Determination of cyanides
DIN EN ISO 10304-1 (D 20) 2009-07	Water quality – Determination of dissolved anions by liquid chromatography of ions – Part 1: Determination of bromide, chloride, fluoride, nitrate, nitrite, phosphate and sulphate (Modification: <i>without bromide and phosphate; additionally: bromate, chlorate and chlorite</i>)
DIN 38405-D 21 1990-10	Determination of dissolved silicate by spectrometry
DIN EN ISO 10304-4 (D 25) 1999-07	Water quality – Determination of dissolved anions by liquid chromatography of ions – Part 4: Determination of chlorate, chloride and chlorite in water with low contamination
DIN EN ISO 15061 (D 34) 2001-12	Water quality – Determination of dissolved bromate – Method by liquid chromatography of ions

1.5 Cations

DIN 38406-E 5 1983-10	Determination of ammonia-nitrogen
DIN EN ISO 17294-2 (E 29) 2024-12	Water quality – Application of inductively coupled plasma mass spectrometry (ICP-MS) – Part 2: Determination of selected elements including uranium isotopes

1.6 Organic parameters

DIN EN ISO 10301 (F 4) 1997-08	Water quality – Determination of highly volatile halogenated hydrocarbons – Gas-chromatographic methods
DIN EN ISO 11369 (F 12) 1997-11	Water quality - Determination of selected plant treatment agents - Method using high performance liquid chromatography with UV detection after solid-liquid extraction (Modification: <i>Measurement with LC-MS/MS</i>)
DIN 38407-F 30 2007-12	Determination of trihalogenmethanes in bathing water and pool water with headspace-gas chromatography

Valid from: 25.08.2025

Date of issue: 25.08.2025

page 4 of 17

This document is a translation. The definitive version is the original German annex to the accreditation certificate.

Annex to the Accreditation Certificate D-PL-20469-01-02

DIN 38407-F 39 2011-09	Determination of selected polycyclic aromatic hydrocarbons (PAH) - Method using gas chromatography with mass spectrometric detection (GC-MS)
DIN EN ISO 20595 2023-08	Water quality - Determination of selected highly volatile organic compounds in water - Method using gas chromatography and mass spectrometry by static headspace technique (HS-GC-MS)

1.7 Gaseous components

DIN EN ISO 7393-2 (G 4-2) 2019-03	Water quality – Determination of free chlorine and total chlorine – Part 2: Colorimetric method using N,N-dialkyl-1,4-phenylenediamine for routine control purposes
DIN EN ISO 5814 (G 22) 2013-02	Water quality – Determination of dissolved oxygen – Electrochemical probe method

1.8 Summary indices of actions and substances

DIN EN ISO 8467 (H 5) 1995-05	Water quality – Determination of permanganate index
DIN 38409-H 6 1986-01	Water hardness
DIN 38409-H 7 2005-12	Determination of acid and base capacity
DIN 38409-H 9 1980-07	Determination of the settleable matter by volume in water and waste water (Restriction: <i>here only determination as per H 9-2 with a sample volume of 2 litres</i>)
DIN ISO 15705 (H 45) 2003-01	Water quality – Determination of the chemical oxygen demand index (ST-COD) – Small-scale sealed tube method

1.9 Quick test with finished reagents for water analysis

Macherey-Nagel GmbH & Co. KG visocolor®ECO Chlor 2 REF 931015 2016-04	Colorimetric determination of free chlorine, total chlorine and bound chlorine in drinking water, swimming pools and water reservoirs using a test kit, measuring range 0.10 - 2.00 mg/L Cl ₂ (Modification: <i>also for industrial water</i>)
--	---

Valid from: 25.08.2025

Date of issue: 25.08.2025

page 5 of 17

This document is a translation. The definitive version is the original German annex to the accreditation certificate.

Annex to the Accreditation Certificate D-PL-20469-01-02

1.10 Microbiological analysis

DIN EN ISO 6222 (K 5) 1999-07	Water quality – Enumeration of culturable micro-organisms – Colony count by inoculation in a nutrient agar culture medium
DIN EN ISO 9308-2 (K 6-1) 2014-06	Water quality – Enumeration of Escherichia coli and coliform bacteria – Part 2: Most probable number method
DIN EN ISO 16266 (K 11) 2008-05	Water quality – Detection and enumeration of Pseudomonas aeruginosa – Membrane filtration method (Modification: <i>Differentiation also with MALDI-TOF-MS</i>)
DIN EN ISO 9308-1 (K 12) 2017-09	Water quality – Enumeration of Escherichia coli and coliform bacteria – Part 1: Membrane filtration method for waters with low bacterial background flora
DIN EN ISO 7899-2 (K 15) 2000-11	Water quality – Detection and enumeration of intestinal enterococci – Part 2: Membrane filtration method
DIN EN ISO 10705-2 (K 17) 2002-01	Water quality – Detection and enumeration of bacteriophages Part 2: Enumeration of somatic coliphages (Modification: <i>here different host strain</i>)
DIN EN ISO 11731 (K 23) 2019-03	Water quality – Enumeration of legionella (Modification: <i>Differentiation also with MALDI-TOF-MS</i>)
DIN EN ISO 14189 (K 24) 2016-11	Water quality – Enumeration of Clostridium perfringens – Method using membrane filtration
UBA Recommendation 2018-12 Amended 2022-12	Systemic analysis of drinking water installations for legionella in accordance with the German Drinking Water Ordinance – Sampling, examination and indication of the result
TrinkwV Annex 1 Section 4 (a) 05 December 1990	Microbiological test methods – Sulphite-reducing spore-forming anaerobes – Examination by liquid enrichment
TrinkwV Section 43 (3)	Colony count at 22°C and 36°C in drinking water by the pour plate technique

Valid from: 25.08.2025

Date of issue: 25.08.2025

page 6 of 17

This document is a translation. The definitive version is the original German annex to the accreditation certificate.

Annex to the Accreditation Certificate D-PL-20469-01-02
2 Test in accordance with the German Drinking Water Ordinance – TrinkwV -
 Drinking Water Ordinance (TrinkwV) of 20 June 2023 (BGBl. 2023 I No. 159, p. 2)

Sampling

Method	Title
DIN ISO 5667-5 2011-02	Water quality – Sampling – Part 5: Guidance on sampling of drinking water from treatment works and piped distribution systems
DIN EN ISO 19458 2006-12	Water quality – Sampling for microbiological analysis
UBA Recommendation 18 December 2018 (legionella)	Systemic examination of drinking water installations for Legionella in accordance with the German Drinking Water Regulation – Sampling, examination and indication of the result
Recommendation of the Federal Environment Agency 18 December 2018 (staggered stagnation sampling and random sampling)	Assessment of the quality of drinking water with respect to the parameters lead, copper and nickel

ANNEX 1: MICROBIOLOGICAL PARAMETERS
Part I: General requirements for drinking water

Parameter	Method
Escherichia coli (E. coli)	DIN EN ISO 9308-1 2017-09 ----- DIN EN ISO 9308-2 2014-06
Intestinal enterococci	DIN EN ISO 7899-2 2000-11

PART II: Requirements for drinking water intended for transfer in sealed containers

Parameter	Method
Escherichia coli (E. coli)	not used
Intestinal enterococci	not used
Pseudomonas aeruginosa	DIN EN ISO 16266 2008-05

ANNEX 2: CHEMICAL PARAMETERS
Part I: Chemical parameters whose concentration does not usually increase in the distribution network, including the drinking water installation

Parameter	Method
Acrylamide	not used
Benzene	DIN EN ISO 20595 2023-08

Valid from: 25.08.2025

Date of issue: 25.08.2025

page 7 of 17

This document is a translation. The definitive version is the original German annex to the accreditation certificate.

Annex to the Accreditation Certificate D-PL-20469-01-02

Parameter	Method
Boron	DIN EN ISO 17294-2 2024-12
Bromate	DIN EN ISO 10304-1 2009-07 (Modification: <i>here also bromate</i>)
Chromium	DIN EN ISO 17294-2 2024-12
Cyanide	DIN 38405-13 2011-04
1,2-dichloroethane	DIN EN ISO 20595 2023-08
Fluoride	DIN EN ISO 10304-1 2009-07
Microcystin-LR	not used
Nitrate	DIN 38405-9 2011-09 DIN EN ISO 10304-1 2009-07
Pesticides	DIN EN ISO 11369 1997-11 (Modification: <i>Measurement with LC-MS/MS</i>)
Total pesticides	DIN EN ISO 11369 1997-11 (Modification: <i>Measurement with LC-MS/MS</i>)
Total PFAS-20	not used
Total PFAS-4	not used
Mercury	DIN EN ISO 17294-2 2024-12
Selenium	DIN EN ISO 17294-2 2024-12
Tetrachloroethylene and trichloroethylene	DIN EN ISO 10301 1997-08
Uranium	DIN EN ISO 17294-2 2024-12

Part II: Chemical parameters whose concentration may increase in the distribution network, including the drinking water installation

Parameter	Method
Antimony	DIN EN ISO 17294-2 2024-12
Arsenic	DIN EN ISO 17294-2 2024-12
Benzo(a)pyrene	DIN 38407-39 2011-09
Bisphenol A	not used
Lead	DIN EN ISO 17294-2 2024-12
Cadmium	DIN EN ISO 17294-2 2024-12
Chlorate	DIN EN ISO 10304-1 2009-07 (Modification: <i>here also chlorate</i>)
	DIN EN ISO 10304-4 1999-07
Chlorite	DIN EN ISO 10304-1 2009-07 (Modification: <i>here also chlorite</i>)
	DIN EN ISO 10304-4 1999-07

Valid from: 25.08.2025

Date of issue: 25.08.2025

page 8 of 17

This document is a translation. The definitive version is the original German annex to the accreditation certificate.

Annex to the Accreditation Certificate D-PL-20469-01-02

Parameter	Method
Epichlorohydrin	not used
Haloacetic acids (HAA-5)	not used
Copper	DIN EN ISO 17294-2 2024-12
Nickel	DIN EN ISO 17294-2 2024-12
Nitrite	DIN EN 26777-10 1993-04
	DIN EN ISO 10304-1 2009-07
Polycyclic aromatic hydrocarbons (PAH)	DIN 38407-39 2011-09
Trihalomethanes (THM)	DIN EN ISO 10301 1997-08
Vinyl chloride	not used

ANNEX 3: INDICATOR PARAMETERS
Part I: General indicator parameters

Parameter	Method
Aluminium	DIN EN ISO 17294-2 2024-12
Ammonium	DIN 38406-5 1983-10
Calcite dissolving capacity	DIN 38404-10 2012-12
Chloride	DIN EN ISO 10304-1 2009-07
	DIN EN ISO 10304-4 1999-07
Clostridium perfringens, including spores	DIN EN ISO 14189 2016-11
Coliform bacteria	DIN EN ISO 9308-1 2017-09
	DIN EN ISO 9308-2 2014-06
Iron	DIN EN ISO 17294-2 2024-12
Electrical conductivity	DIN EN 27888 1993-11
Colouring	DIN EN ISO 7887 2012-04
Odour	DIN EN 1622 2006-10 (Annex C)
Flavour	DIN EN 1622 2006-10 (Annex C)
Colony count at 22 °C	DIN EN ISO 6222 1999-07
	TrinkwV Section 43 (3)
Colony count at 36 °C	DIN EN ISO 6222 1999-07
	TrinkwV Section 43 (3)
Manganese	DIN EN ISO 17294-2 2024-12
Sodium	DIN EN ISO 17294-2 2024-12
Organically bound carbon (TOC)	not used

Valid from: 25.08.2025

Date of issue: 25.08.2025

page 9 of 17

This document is a translation. The definitive version is the original German annex to the accreditation certificate.

Annex to the Accreditation Certificate D-PL-20469-01-02

Parameter	Method
Oxidisability	DIN EN ISO 8467 1995-05
Sulphate	DIN EN ISO 10304-1 2009-07
Turbidity	DIN EN ISO 7027-1 2016-11
Hydrogen ion concentration	DIN EN ISO 10523 2012-04

Part II: Specific indicator parameter for drinking water installations

Parameter	Method
Legionella spec.	DIN EN ISO 11731 2019-03 UBA Recommendation 18 December 2018 Amended December 2022 (Federal Health Gazette 2023 p. 224)

Part III: Specific indicator parameter for the occurrence of certain microbial hazards

Parameter	Method
Somatic coliphages	DIN EN ISO 10705-2 2002-01 (Modification: <i>here different host strain</i>)

ANNEX 4: REQUIREMENTS FOR DRINKING WATER WITH REGARD TO RADIOACTIVE SUBSTANCES

not used

PARAMETERS NOT INCLUDED IN ANNEXES 1 TO 4 OF THE GERMAN DRINKING WATER ORDINANCE
Additional periodic testing

Parameter	Method
Calcium	DIN EN ISO 17294-2 2024-12
Potassium	DIN EN ISO 17294-2 2024-12
Magnesium	DIN EN ISO 17294-2 2024-12
Acid and base capacity	DIN 38409-7 2005-12
Phosphate	DIN EN ISO 6878 2004-09

The accreditation does not replace the recognition or approval procedure of the competent authority pursuant to Section 40 (2) TrinkwV.

Valid from: 25.08.2025

Date of issue: 25.08.2025

page 10 of 17

This document is a translation. The definitive version is the original German annex to the accreditation certificate.

3 Sampling and microbiological analysis of industrial water in accordance with Section 3 (8) 42nd BImSchV

Sampling

Method	Title
DIN EN ISO 19458 (K 19) 2006-12	Water quality – Sampling for microbiological analysis Recommendation of the Federal Environmental Agency for the sampling and detection of Legionella in evaporative cooling plants, cooling towers and wet separators dated 06.03.2020, Sections C and D

Microbiological analyses

Parameter	Method
Legionella	DIN EN ISO 11731 (K 23) 2019-03 Recommendation of the Federal Environmental Agency for the sampling and detection of Legionella in evaporative cooling plants, cooling towers and wet separators dated 06.03.2020, Sections E and F taking into account Annexes 1 and 2
Colony count at 22 °C and 36 °C	DIN EN ISO 6222 (K 5) 1999-07

4 List of test methods for the SPECIALIST MODULE FOR WATER

Revised: LAWA of 18.10.2018

Section 1: Sampling and general parameters

not used

Section 2: Photometry, ion chromatography, titrimetry

Parameter	Method	Was	Sur	Raw
Absorption at 254 nm (SAC 254)	DIN 38404-C 3: 2005-07	☐	☐	☒
Absorption at 436 nm (SAC 436)	DIN EN ISO 7887: 2012-04 (C 1), Method B	☐	☐	☒

Valid from: 25.08.2025

Date of issue: 25.08.2025

page 11 of 17

This document is a translation. The definitive version is the original German annex to the accreditation certificate.

Annex to the Accreditation Certificate D-PL-20469-01-02

Parameter	Method	Was	Sur	Raw
Ammonium nitrogen	DIN EN ISO 11732: 2005-05 (E 23)	☐	☐	☐
	DIN 38406-E 5: 1983-10	☐	☐	☒
	DIN EN ISO 14911: 1999-12 (E 34)	☐	☐	☐
	DIN ISO 15923-1: 2014-07 (D 49)	☐	☐	☐
Nitrite nitrogen	DIN EN 26777: 1993-04 (D 10)	☐	☐	☐
	DIN EN ISO 10304-1: 2009-07 (D 20)	☐	☐	☒
	DIN EN ISO 13395: 1996-12 (D 28)	☐	☐	☐
	DIN ISO 15923-1: 2014-07 (D 49)	☐	☐	☐
Nitrate nitrogen	DIN EN ISO 10304-1: 2009-07 (D 20)	☐	☐	☒
	DIN EN ISO 13395: 1996-12 (D 28)	☐	☐	☐
	DIN 38405-D 9: 2011-09	☐	☐	☐
	DIN 38405-D 29: 1994-11	☐	☐	☐
	DIN ISO 15923-1: 2014-07 (D 49)	☐	☐	☐
Phosphorus, total (see also section 3)	DIN EN ISO 6878: 2004-09 (D 11)	☐	☐	☐
	DIN EN ISO 15681-1: 2005-05 (D 45)	☐	☐	☐
	DIN EN ISO 15681-2: 2005-05 (D 46)	☐	☐	☐
Orthophosphate	DIN EN ISO 10304-1: 2009-07 (D 20)	☐	☐	☐
	DIN EN ISO 6878: 2004-09 (D 11)	☐	☐	☒
	DIN EN ISO 15681-1: 2004-07 (D 45)	☐	☐	☐
	DIN EN ISO 15681-2: 2005-05 (D 46)	☐	☐	☐
	DIN ISO 15923-1: 2014-07 (D 49)	☐	☐	☐
Fluoride (dissolved)	DIN 38405-D 4-1, 1985-07	☐	☐	☐
	DIN EN ISO 10304-1: 2009-07 (D 20)	☐	☐	☐
Chloride	DIN EN ISO 10304-1: 2009-07 (D 20)	☐	☐	☒
	DIN EN ISO 15682: 2002-01 (D 31)	☐	☐	☐
	DIN ISO 15923-1: 2014-07 (D 49)	☐	☐	☐
	DIN EN ISO 10304-4: 1999-07 (D 25)	☐	☐	☐
	DIN 38405-D 1-1 and D 1-2: 1985-12	☐	☐	☐
	DIN 38405-D 1-3 and D 1-4: 1985-12	☐	☐	☐

Valid from: 25.08.2025

Date of issue: 25.08.2025

page 12 of 17

This document is a translation. The definitive version is the original German annex to the accreditation certificate.

Annex to the Accreditation Certificate D-PL-20469-01-02

Parameter	Method	Was	Sur	Raw
Sulphate	DIN EN ISO 10304-1: 2009-07 (D 20)	☐	☐	☒
	DIN 38405-D 5-1: 1985-01	☐	☐	☐
	DIN 38405 D 5-2:1985-01	☐	☐	☐
	DIN ISO 15923-1: 2014-07 (D 49)	☐	☐	☐
Cyanide (readily liberated)	DIN 38405-D 13-2: 1981-02	☐	☐	☐
	DIN EN ISO 14403-1: 2012-10 (D 2)	☐	☐	☐
	DIN EN ISO 14403-2: 2012-10 (D 3)	☐	☐	☐
	DIN 38405-D 7: 2002-04	☐	☐	☐
Cyanide (total)	DIN 38405-D 13-1: 1981-02	☐	☐	☐
	DIN EN ISO 14403-1: 2012-10 (D 2)	☐	☐	☐
	DIN EN ISO 14403-2: 2012-10 (D 3)	☐	☐	☐
	DIN 38405-D 7: 2002-04	☐	☐	☐
Chromium(VI)	DIN 38405-D 24: 1987-05	☐	☐	☐
	DIN EN ISO 10304-3: 1997-11 (D 22), Section 6 (dissolved chromate)	☐	☐	☐
	DIN EN ISO 23913: 2009-09 (D 41)	☐	☐	☐
	DIN EN ISO 18412: 2007-02 (D 40)	☐	☐	☐
Sulphide (readily liberated)	DIN 38405-D 27: 1992-07	☐	☐	☐

Section 3: Elemental analysis

Parameter	Method	Was	Sur	Raw
Aluminium	DIN EN ISO 11885: 2009-09 (E 22)	☐	☐	☐
	DIN EN ISO 12020: 2000-05 (E 25)	☐	☐	☐
	DIN EN ISO 17294-2: 2017-01 (E 29)	☐	☐	☒
	DIN EN ISO 15586: 2004-02 (E 4)	☐	☐	☐
Arsenic	DIN EN ISO 11969: 1996-11 (D 18)	☐	☐	☐
	DIN EN ISO 11885: 2009-09 (E 22)	☐	☐	☐
	DIN EN ISO 17294-2: 2017-01 (E 29)	☐	☐	☒
	DIN EN ISO 15586: 2004-02 (E 4)	☐	☐	☐
	DIN 38405-D 35: 2004-09	☐	☐	☐

Valid from: 25.08.2025

Date of issue: 25.08.2025

page 13 of 17

This document is a translation. The definitive version is the original German annex to the accreditation certificate.

Annex to the Accreditation Certificate D-PL-20469-01-02

Parameter	Method	Was	Sur	Raw
Lead	DIN EN ISO 11885: 2009-09 (E 22)	☐		
	DIN 38406-E 6: 1998-07	☐	☐	☐
	DIN EN ISO 17294-2: 2017-01 (E 29)	☐	☐	☐
	DIN EN ISO 15586: 2004-02 (E 4)	☐	☐	☐
Cadmium	DIN EN ISO 11885: 2009-09 (E 22)	☐		
	DIN EN ISO 5961: 1995-05 (E 19)	☐	☐	☐
	DIN EN ISO 17294-2: 2017-01 (E 29)	☐	☐	☐
	DIN EN ISO 15586: 2004-02 (E 4)	☐	☐	☐
Calcium	DIN EN ISO 11885: 2009-09 (E 22)		☐	☐
	DIN 38406-E 3: 2002-03		☐	☐
	DIN EN ISO 7980: 2000-07 (E 3a)		☐	☐
	DIN EN ISO 17294-2: 2017-01 (E 29)		☐	☒
	DIN EN ISO 14911: 1999-12 (E 34)		☐	☐
Chromium	DIN EN ISO 11885: 2009-09 (E 22)	☐	☐	☐
	DIN EN 1233: 1996-08 (E 10)	☐	☐	☐
	DIN EN ISO 17294-2: 2017-01 (E 29)	☐	☐	☐
	DIN EN ISO 15586: 2004-02 (E 4)	☐	☐	☐
Iron	DIN EN ISO 11885: 2009-09 (E 22)	☐	☐	☐
	DIN 38406-E 32: 2000-05	☐	☐	☐
	DIN EN ISO 15586: 2004-02 (E 4)	☐	☐	☐
	DIN EN ISO 17294-2: 2017-01 (E 29)	☐	☐	☒
Potassium	DIN 38406-E 13: 1992-07		☐	☐
	DIN EN ISO 11885: 2009-09 (E 22)		☐	☐
	DIN EN ISO 17294-2: 2017-01 (E 29)		☐	☒
	DIN EN ISO 14911: 1999-12 (E 34)		☐	☐
Copper	DIN EN ISO 11885: 2009-09 (E 22)	☐	☐	☐
	DIN 38406-E 7: 1991-09	☐	☐	☐
	DIN EN ISO 17294-2: 2017-01 (E 29)	☐	☐	☐
	DIN EN ISO 15586: 2004-02 (E 4)	☐	☐	☐

Valid from: 25.08.2025

Date of issue: 25.08.2025

page 14 of 17

This document is a translation. The definitive version is the original German annex to the accreditation certificate.

Annex to the Accreditation Certificate D-PL-20469-01-02

Parameter	Method	Was	Sur	Raw
Manganese	DIN EN ISO 11885: 2009-09 (E 22)	☐	☐	☐
	DIN EN ISO 17294-2: 2017-01 (E 29)	☐	☐	☒
	DIN 38406-E 33: 2000-06	☐	☐	☐
	DIN EN ISO 15586: 2004-02 (E 4)	☐	☐	☐
	DIN EN ISO 14911: 1999-12 (E 34)	☐	☐	☐
Sodium	DIN 38406-E 14: 1992-07	☐	☐	☐
	DIN EN ISO 11885: 2009-09 (E 22)	☐	☐	☐
	DIN EN ISO 17294-2: 2017-01 (E 29)	☐	☐	☒
	DIN EN ISO 14911: 1999-12 (E 34)	☐	☐	☐
Nickel	DIN EN ISO 11885: 2009-09 (E 22)	☐	☐	☐
	DIN 38406-E 11: 1991-09	☐	☐	☐
	DIN EN ISO 17294-2: 2017-01 (E 29)	☐	☐	☐
	DIN EN ISO 15586: 2004-02 (E 4)	☐	☐	☐
Mercury	DIN EN ISO 17852: 2008-04 (E 35)	☐	☐	☐
	DIN EN ISO 12846: 2012-08 (E 12)	☐	☐	☐
Zinc	DIN EN ISO 11885: 2009-09 (E 22)	☐	☐	☐
	DIN 38406-E 8: 2004-10	☐	☐	☐
	DIN EN ISO 17294-2: 2017-01 (E 29)	☐	☐	☐
	DIN EN ISO 15586: 2004-02 (E 4)	☐	☐	☐
Boron	DIN EN ISO 11885: 2009-09 (E 22)	☐	☐	☐
	DIN EN ISO 17294-2: 2017-01 (E 29)	☐	☐	☐
Magnesium	DIN EN ISO 11885: 2009-09 (E 22)	☐	☐	☐
	DIN 38406-E 3: 2002-03	☐	☐	☐
	DIN EN ISO 7980: 2000-07 (E 3a)	☐	☐	☐
	DIN EN ISO 17294-2: 2017-01 (E 29)	☐	☐	☒
	DIN EN ISO 14911: 1999-12 (E 34)	☐	☐	☐
Phosphorus, total (see also section 2)	DIN EN ISO 11885: 2009-09 (E 22)	☐	☐	☐
	DIN EN ISO 17294-2: 2017-01 (E 29)	☐	☐	☐

Valid from: 25.08.2025

Date of issue: 25.08.2025

page 15 of 17

This document is a translation. The definitive version is the original German annex to the accreditation certificate.

Section 4/5: Group and sum parameters

Parameter	Method	Was	Sur	Raw
Biological oxygen demand (BOD ₅)	DIN EN 1899-1: 1998-05 (H 51)	☐	☐	☐
	DIN EN 1899-2: 1998-05 (H 52)	☐	☐	☐
Chemical oxygen demand (COD)	DIN 38409-H 41: 1980-12	☐	☐	☐
	DIN 38409-H 44: 1992-05	☐	☐	☐
	DIN ISO 15705: 2003-01 (H 45)	☐	☐	☐
Phenol index	DIN 38409-H 16-2: 1984-06	☐	☐	☐
	DIN 38409-H 16-1: 1984-06	☐	☐	☐
	DIN EN ISO 14402: 1999-12 (H 37) Method as per section 4	☐	☐	☐
Filterable solids	DIN EN 872: 2005-04 (H 33)	☐	☐	☐
	DIN 38409-H 2-3: 1987-03	☐	☐	☐
Acid and base capacity	DIN 38409-H 7: 2005-12	☐	☐	☒
Total organic carbon (TOC)	DIN EN 1484: 1997-08 (H 3)	☐	☐	☐
Dissolved organic carbon (DOC)	DIN EN 1484: 1997-08 (H 3)	☐	☐	☐
Total bound nitrogen (TN _b)	DIN EN 12260: 2003-12 (H 34)	☐	☐	☐
	DIN EN ISO 11905-1: 1998-08 (H 36)	☐	☐	☐
Adsorbable organic halogens (AOX)	DIN EN ISO 9562: 2005-02 (H 14)	☐	☐	☐

Section 6: Gas chromatographic methods

not used

Section 7: HPLC methods

not used

Section 8: Microbiological methods (not used)
Section 9.1: Biological methods, bio-assays (part 1)

not used

Section 9.2: Biological methods, bio-assays (part 2)

not used

Valid from: 25.08.2025

Date of issue: 25.08.2025

Annex to the Accreditation Certificate D-PL-20469-01-02

Abbreviations used:

DEV	Deutsche Einheitsverfahren zur Wasser-, Abwasser- und Schlammuntersuchung (German standard methods for analysis of water, waste water and sludge)
DIN	Deutsches Institut für Normung e.V. (German Institute for Standardization)
DMA	Direct mercury analyzer
EN	Europäische Norm (European standard)
IEC	International Electrotechnical Commission
ISO	International Organization for Standardization
MUVA-MET	In-house method of muva kempten GmbH
UBA	Umweltbundesamt (Federal Environment Agency)

Valid from: 25.08.2025

Date of issue: 25.08.2025

page 17 of 17

This document is a translation. The definitive version is the original German annex to the accreditation certificate.